

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T13 · Ratio Problem Solving

With Answers

Questions with full worked solutions

■ 27 pages

FAMILY MAP

Ratio Problem Solving

F01	Update shares after allocation	2 Q 8 marks
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F02	Recover the ratio scale from a difference	6 Q 23 marks
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F03	Recover the ratio scale from partial information	4 Q 16 marks
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6 Pau, Sam and Tia share £240 in the ratios 3 : 4 : 5

Sam and Tia each give £10 of their share to Pau.

Work out the ratios of the amounts of money that Pau, Sam and Tia now have.
Give your answer in its simplest form.

..... : :

(Total for Question 6 is 4 marks)

Worked Solution 21

Ratio after transfers

UNIT 2
4 marks**Initial shares**

$$240 \div (3 + 4 + 5) = 20 \text{ per part; } (P, S, T) = (60, 80, 100).$$

Transfers

$$(P, S, T) = (60 + 10 + 10, 80 - 10, 100 - 10) = (80, 70, 90).$$

Simplify

$$80 : 70 : 90 = 8 : 7 : 9.$$

Final answer

$$8 : 7 : 9$$

Question 06

4MA1-1H Q14

ORIGINAL QUESTION

5 marks

14 The combined savings of Abel and Bahira are 15 435 dinars.

The savings of Bahira are 45% more than the savings of Abel.

The savings of Bahira are $\frac{3}{2}$ times the savings of Chanda.

Work out the savings of Chanda.

..... dinars

Worked Solution 06

Linked ratio

MODULAR UNIT 2

5 marks

Abel and Bahira

$$B = 1.45A, \quad A + B = 15435 \Rightarrow A = 6300, \quad B = 9135.$$

Chanda

$$B = \frac{3}{2}C \Rightarrow C = \frac{2}{3}(9135) = 6090.$$

Final answer

6090 dinars

Question 14

4MA1-2H Q8

ORIGINAL QUESTION

5 marks

- 8 Ishir plants 600 bulbs in a garden.
He plants tulip bulbs, crocus bulbs and daffodil bulbs so that

number of tulip bulbs : number of crocus bulbs : number of daffodil bulbs = 9 : 4 : 2

45% of the tulip bulbs are for yellow flowers.

$\frac{5}{8}$ of the crocus bulbs are for yellow flowers.

All of the daffodil bulbs are for yellow flowers.

Work out the number of bulbs that are for yellow flowers.

Worked Solution 14

Ratio and percentages

MODULAR UNIT 2

5 marks

Bulb counts

$$600/15 = 40; \quad T = 360, \quad C = 160, \quad D = 80.$$

Yellow

$$0.45(360) + \frac{5}{8}(160) + 80 = 342.$$

Final answer

342 bulbs

Question 01

4MA1-1H Q1

ORIGINAL QUESTION

3 marks

- 1 Last season, the number of goals scored by Arjun, by Simon and by Kath for their football team were in the ratios $2:5:8$

Simon scored 12 more goals than Arjun.

Work out the number of goals scored by Kath.

Worked Solution 01

Ratio difference

MODULAR UNIT 2

3 marks

Find one ratio part

$$5 - 2 = 3 \text{ parts} = 12 \Rightarrow 1 \text{ part} = 4.$$

Find Kath's goals

$$8 \times 4 = 32.$$

Final answer

32

Original question 02

4MA1-1H Q2

ORIGINAL QUESTION**3 marks**

- 2 Danil, Gabriel and Hadley share some money in the ratios 3 : 5 : 9
The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros.

Work out the amount of money that Danil receives.

..... euros

Worked solution 02

Chapter: Ratio Problem Solving

MODULAR UNIT 2

3 marks

Q2 – Chapter: Ratio Problem Solving

Family: Recover the ratio scale from a difference • Variant: Difference between two shares

Method

The difference in shares is $(9-5)=4$ parts, so one part is $196 \div 4 = 49$.

Why: Gabriel and Hadley correspond to 5 and 9 parts.

Method

Danil receives $3 \times 49 = 147$.

Why: Danil has three ratio parts.

Method

Therefore Danil receives 147 euros.

Why: Give the requested amount with units.

Final answer

Danil: 147 euros

- 5 120 children go on an activity holiday.
The ratio of the number of girls to the number of boys is 3 : 5
- On Sunday, all the children either go sailing or go climbing.
- $\frac{16}{25}$ of the boys go climbing.
- Twice as many girls go sailing as go climbing.
- Work out how many children go sailing on Sunday.

Worked solution

January 2020 | Question 5 | 6 marks

Family: Recover the ratio scale from partial information • Variant: Known fraction or remainder then ratio split

Method / working

1. 3:5 gives 45 girls and 75 boys.
2. Boys climbing= $(16/25) \times 75=48$.
3. Boys sailing= $75-48=27$.
4. Girls sailing:climbing=2:1, so girls sailing= $(2/3) \times 45=30$.
5. Total sailing= $27+30$.
6. Answer=57 children.

Final answer: 57 children sailing

- 1 Pieter owns a currency conversion shop.

Last Monday, Pieter changed a total of 20 160 rand into a number of different currencies.

He changed $\frac{3}{10}$ of the 20 160 rand into euros.

He changed the rest of the rands into dollars, rupees and francs in the ratios 9 : 5 : 2

Pieter changed more rands into dollars than he changed into francs.

Work out how many more.

..... rand

Worked solution

January 2021 | Question 1 | 4 marks

Family: Recover the ratio scale from partial information • Variant: Known fraction or remainder then ratio split

Method / working

1. Euros = $\frac{3}{10} \times 20160 = 6048$, so the remainder is 14112.
2. The ratio total is $9+5+2=16$, giving one part $14112/16=882$.
3. Dollars minus francs = $(9-2) \times 882$.
4. Therefore the difference is 6174 rand.

Final answer: 6174 rand

10 The people working for a company work in Team A or in Team B.

number of people in Team A : number of people in Team B = 3 : 4

$\frac{4}{5}$ of Team A work full time.

24% of Team B work full time.

Work out what fraction of the people working for the company work full time.
Give your fraction in its simplest form.

Worked solution

January 2021 | Question 10 | 3 marks

Family: Recover the ratio scale from partial information • Variant: Known combined share

Method / working

1. Let Team A=3k and Team B=4k.
 2. Full-time fraction= $(\frac{4}{5} \times 3k + 0.24 \times 4k) / (7k)$.
 3. Simplify to 12/25.
- Final answer: 12/25

- 4 On a farm there are chickens, ducks and pigs.

The ratio of the number of chickens to the number of ducks is 7:2

The ratio of the number of ducks to the number of pigs is 5:9

There are 36 pigs on the farm.

Work out the number of chickens on the farm.

Worked solution

June 2021 | Question 4 | 3 marks

Family: Recover the ratio scale from a difference • Variant: Difference after linked or weighted ratios

Method / working

1. 36 pigs correspond to 9 ratio parts, so one part is 4 and ducks = $5 \times 4 = 20$.

2. Chickens : ducks = $7 : 2$, so chickens = $20 \times 7 / 2$.

3. Therefore the number of chickens is 70.

Final answer: 70 chickens

- 4 Andreas, Isla and Paulo share some money in the ratios 3 : 2 : 5

The **total** amount of money that Isla and Paulo receive is £76 more than the amount of money that Andreas receives.

Andreas buys a video game for £48.50 with some of his share of the money.

Work out how much money Andreas has left from his share of the money when he has bought the video game.

£.....

(Total for Question 4 is 4 marks)

Answer reference | May-Nov 2020 | Question 4

Family: Recover the ratio scale from a difference • Variant: Difference between two shares

4	eg $76 \div (5 + 2 - 3)$ oe (= 19) or $5x + 2x - 3x = 76$ and $x = 76 \div (5 + 2 - 3)$ (=19) oe			M1 For a correct method to find the value of 1 share
	$3 \times "19" (= 57)$			M1
	"57" – 48.5(0)			M1
		8.5(0)	4	A1
				Total 4 marks

Worked solution

May-Nov 2020 | Question 4 | 4 marks

Family: Recover the ratio scale from a difference • Variant: Difference between two shares

Official final mark-scheme pages are embedded immediately after this question.

- 5 In a box, there are only green sweets, orange sweets and yellow sweets.

There are 280 sweets in the box so that

the number of green sweets : the number of orange sweets = 2 : 3

and

the number of orange sweets : the number of yellow sweets = 1 : 5

Work out how many green sweets there are in the box.

(Total for Question 5 is 3 marks)

Answer reference | January 2023 | Question 5

Family: Recover the ratio scale from partial information • Variant: Known fraction or remainder then ratio split

5	2 : 3 : 15 oe or 20 or (1 : 5) × 3 or (1 : 5 ⇒) 3 : 15 or 2n : 3n : 15n e.g. 4 : 6 : 30 or G(reen) = 2, O(range) = 3, Y(ellow) = 15		3	M1
	$\frac{2}{20} \cdot 280$ oe or 14×2 or $\frac{2}{2+3+15} \cdot 280$ oe or $\frac{2n}{2n+3n+15n} \cdot 280$ oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	28		A1 or 28 : 42 : 210 or 28, 42, 210 If not in this order must be labelled correctly
				Total 3 marks

Worked solution

January 2023 | Question 5 | 3 marks

Family: Recover the ratio scale from partial information • Variant: Known fraction or remainder then ratio split

Official final mark-scheme pages are embedded immediately after this question.

6 Eli and Peta share \$275 in the ratio 2 : 3

Eli gives $\frac{3}{11}$ of his share to charity.

Peta gives 0.32 of her share to charity.

Work out the total amount that Eli and Peta give to charity.

\$.....

(Total for Question 6 is 4 marks)

Worked solution

November 2025 | Question 6 | 4 marks

Family: Update shares after allocation • Variant: Fixed transfers after sharing

Method

The ratio is

Eli : Peta = 2:3

There are 5 parts, so one part is

$275 / 5 = 55$

Eli gets

$2 \times 55 = 110$

Peta gets

$3 \times 55 = 165$

Eli gives

$(3)/(11) \times 110 = 30$

Peta gives

$0.32 \times 165 = 52.8$

Total given to charity:

$30 + 52.8 = 82.8$

Answer: 82.8.